Literal Equations & Formulas

Lesson 2-5 · Rewriting Formulas / Solving for a Variable

Name: _____ **Date:** _____

A **literal equation** is an equation with more than one variable — usually a **formula**. To *rewrite* (or *solve*) it for one variable, isolate that variable with inverse operations, treating every other letter as if it were a constant. Difficulty tags: [Easy] [Medium] [Hard].

1. Circumference of a Circle

$C = \pi d = 2\pi r$. Solve for the radius: $r = \dfrac{C}{2\pi}$; solve for the diameter: $d = \dfrac{C}{\pi}$.

Detailed Example. Solve $C = 2\pi r$ for r, then find r when $C = 2\pi$.

$$C = 2\pi r \qquad \text{original formula}$$
$$\frac{C}{2\pi} = r \qquad \text{divide both sides by } 2\pi$$
$$r = \frac{C}{2\pi} = \frac{2\pi}{2\pi} = 1 \qquad \text{substitute } C = 2\pi \text{ and simplify}$$

1. [Easy] Solve $C = 2\pi r$ for r.

2. [Easy] A circle has circumference $C = 12\pi$. Find r.

3. [Medium] Solve $C = \pi d$ for d, then find d when $C = 15\pi$.

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4. [Medium] Find r when $C = 9\pi$.

5. [Hard] A circle has circumference 31.4. Find r (use $\pi \approx 3.14$).

6. [Hard] Solve $C = 2\pi r$ for r, then find r when $C = 100$ ($\pi \approx 3.14$, round to a tenth).

2. Area of a Rectangle

$A = \ell w$. Solve for width: $w = \dfrac{A}{\ell}$; solve for length: $\ell = \dfrac{A}{w}$.

Detailed Example. Solve $A = \ell w$ for w, then find w when $A = 12$ and $\ell = 2$.

$A = \ell w$	original formula
$\dfrac{A}{\ell} = w$	divide both sides by ℓ (treat A as a constant)
$w = \dfrac{A}{\ell} = \dfrac{12}{2} = 6$	substitute and simplify

1. [Easy] Solve $A = \ell w$ for w.

2. [Easy] Solve $A = \ell w$ for ℓ.

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3. [Medium] Find w when $A = 48$ and $\ell = 6$.

4. [Medium] A rectangle has area 54 and length 9. Find the width.

5. [Hard] Find ℓ when $A = \frac{3}{4}$ and $w = \frac{1}{2}$.

6. [Hard] A rectangle has area 2.4 m^2 and length 0.8 m. Find the width.

3. Area of a Triangle

$A = \frac{1}{2}bh$. Solve for height: $h = \frac{2A}{b}$; solve for base: $b = \frac{2A}{h}$.

Detailed Example. Solve $A = \frac{1}{2}bh$ for h.

$$\begin{aligned} \tfrac{1}{2}bh &= A && \text{original formula (symmetric property)} \\ bh &= 2A && \text{multiply both sides by 2} \\ h &= \frac{2A}{b} && \text{divide both sides by } b \end{aligned}$$

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1. [Easy] Solve $A = \frac{1}{2}bh$ for h.

2. [Easy] Solve $A = \frac{1}{2}bh$ for b.

3. [Medium] Find h when $A = 24$ and $b = 6$.

4. [Medium] A triangle has area 30 and height 5. Find the base.

5. [Hard] Find h when $A = 15$ and $b = 2.5$.

6. [Hard] Find h when $A = \frac{3}{8}$ and $b = \frac{3}{4}$.

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4. Perimeter of a Rectangle

$P = 2\ell + 2w$. Solve for width: $w = \dfrac{P - 2\ell}{2}$; solve for length: $\ell = \dfrac{P - 2w}{2}$.

Detailed Example. Solve $P = 2\ell + 2w$ for w.

$$\begin{aligned} 2\ell + 2w &= P && \text{original formula} \\ 2w &= P - 2\ell && \text{subtract } 2\ell \text{ from both sides} \\ w &= \frac{P - 2\ell}{2} && \text{divide both sides by 2} \end{aligned}$$

1. [Easy] Solve $P = 2\ell + 2w$ for w.

2. [Easy] Solve $P = 2\ell + 2w$ for ℓ.

3. [Medium] Find w when $P = 20$ and $\ell = 6$.

4. [Medium] A rectangle has perimeter 30 and width 4. Find the length.

5. [Hard] Find ℓ when $P = 25$ and $w = 3.5$.

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6. [Hard] A rectangle has perimeter 15 and length 4.5. Find the width.

5. Distance = Rate × Time

$d = rt$. Solve for time: $t = \dfrac{d}{r}$; solve for rate: $r = \dfrac{d}{t}$.

Detailed Example. Solve $d = rt$ for t, given $r = 60$.

$$d = rt \qquad \text{original formula}$$
$$\frac{d}{r} = t \qquad \text{divide both sides by } r$$
$$t = \frac{d}{r} = \frac{d}{60} \qquad \text{substitute } r = 60$$

1. [Easy] Solve $d = rt$ for t.

2. [Easy] Solve $d = rt$ for r.

3. [Medium] Find t when $d = 150$ and $r = 50$.

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4. [Medium] A car travels 240 mi at 60 mph. Find the time.

5. [Hard] Find r when $d = 0.5$ mi and $t = 0.25$ h.

6. [Hard] A runner covers 13.1 mi in 2 h. Find the average rate (round to a tenth).

6. Temperature Conversion

$C = \frac{5}{9}(F - 32)$. To solve for F: multiply by $\frac{9}{5}$, then add 32: $F = \frac{9}{5}C + 32$.

Detailed Example. If $F = 77°$, find C; then if $C = 25°$, solve for F.

$$C = \tfrac{5}{9}(77 - 32) = \tfrac{5}{9}(45) = 25° \quad \text{substitute } F = 77$$
$$25 = \tfrac{5}{9}(F - 32) \quad \text{now solve for } F$$
$$25 \cdot \tfrac{9}{5} = F - 32 \;\Rightarrow\; 45 = F - 32 \quad \text{multiply both sides by } \tfrac{9}{5}$$
$$F = 77° \quad \text{add 32 to both sides}$$

1. [Easy] Find C when $F = 41°$.

2. [Easy] Find C when $F = 59°$.

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3. [Medium] Solve $C = \frac{5}{9}(F - 32)$ for F.

4. [Medium] Find F when $C = 100°$.

5. [Hard] Find F when $C = 37°$ (round to a tenth).

6. [Hard] Find C when $F = 98.6°$.

7. General Literal Equations

To solve a multi-variable equation for one variable, treat the other letters as constants. Undo addition/subtraction first, then multiplication/division. Recall the *symmetric property*: if $a = b$, then $b = a$.

Detailed Example. Solve $4 = 2m - 5n$ for m, then evaluate at $n = -2$.

$$\begin{aligned} 2m - 5n &= 4 && \text{symmetric property of equality} \\ 2m &= 4 + 5n && \text{add } 5n \text{ to both sides} \\ m &= \frac{4 + 5n}{2} && \text{divide both sides by 2} \\ m &= \frac{4 + 5(-2)}{2} = \frac{-6}{2} = -3 && \text{substitute } n = -2 \end{aligned}$$

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1. [Easy] Solve $y = x + 7$ for x.

2. [Easy] Solve $3x = y$ for x.

3. [Medium] Solve $2m - 5n = 4$ for m.

4. [Medium] Solve $ax + b = c$ for x.

5. [Hard] Solve $3p - 2q = 12$ for q, then find q when $p = 6$.

6. [Hard] Solve $5a + 2b = 20$ for b, then find b when $a = 2$.

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8. Word Problems

1. [Easy] The circumference of a circle is $C = 2\pi r$. A circular garden has circumference 18π ft. What is its radius?

2. [Medium] The area of a rectangular rug is 24 ft^2 and its length is 6 ft. Rewrite $A = \ell w$ to solve for width, then find the width.

3. [Medium] The distance formula is $d = rt$. A train travels 330 miles in 5.5 hours. Rewrite the formula to solve for rate, then find the average rate.

4. [Hard] The formula $C = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius.

   (a) Rewrite the formula to solve for F. (b) A recipe says to bake at 190°C. What is that in °F?

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9. Detailed Answer Key

1. Circumference of a Circle

1. $r = \dfrac{C}{2\pi}$.
2. $r = \dfrac{12\pi}{2\pi} = 6$.
3. $d = \dfrac{C}{\pi}$; $d = \dfrac{15\pi}{\pi} = 15$.
4. $r = \dfrac{9\pi}{2\pi} = 4.5$.
5. $r = \dfrac{31.4}{2(3.14)} = \dfrac{31.4}{6.28} = 5$.
6. $r = \dfrac{C}{2\pi} = \dfrac{100}{6.28} \approx 15.9$.

2. Area of a Rectangle

1. $w = \dfrac{A}{\ell}$.
2. $\ell = \dfrac{A}{w}$.
3. $w = \dfrac{48}{6} = 8$.
4. $w = \dfrac{54}{9} = 6$.
5. $\ell = \dfrac{3/4}{1/2} = \dfrac{3}{4} \cdot 2 = \dfrac{3}{2}$.
6. $w = \dfrac{2.4}{0.8} = 3$ m.

3. Area of a Triangle

1. $h = \dfrac{2A}{b}$.
2. $b = \dfrac{2A}{h}$.
3. $h = \dfrac{2(24)}{6} = 8$.
4. $b = \dfrac{2(30)}{5} = 12$.
5. $h = \dfrac{2(15)}{2.5} = \dfrac{30}{2.5} = 12$.

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6. $h = \dfrac{2(3/8)}{3/4} = \dfrac{3/4}{3/4} = 1.$

4. Perimeter of a Rectangle

1. $w = \dfrac{P - 2\ell}{2}.$
2. $\ell = \dfrac{P - 2w}{2}.$
3. $w = \dfrac{20 - 12}{2} = 4.$
4. $\ell = \dfrac{30 - 8}{2} = 11.$
5. $\ell = \dfrac{25 - 7}{2} = 9.$
6. $w = \dfrac{15 - 9}{2} = 3.$

5. Distance = Rate × Time

1. $t = \dfrac{d}{r}.$
2. $r = \dfrac{d}{t}.$
3. $t = \dfrac{150}{50} = 3.$
4. $t = \dfrac{240}{60} = 4$ h.
5. $r = \dfrac{0.5}{0.25} = 2$ mph.
6. $r = \dfrac{13.1}{2} = 6.55 \approx 6.6$ mph.

6. Temperature Conversion

1. $C = \frac{5}{9}(41 - 32) = \frac{5}{9}(9) = 5°.$
2. $C = \frac{5}{9}(59 - 32) = \frac{5}{9}(27) = 15°.$
3. $F = \frac{9}{5}C + 32.$
4. $F = \frac{9}{5}(100) + 32 = 180 + 32 = 212°.$
5. $F = \frac{9}{5}(37) + 32 = 66.6 + 32 = 98.6°.$
6. $C = \frac{5}{9}(98.6 - 32) = \frac{5}{9}(66.6) = 37°.$

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7. General Literal Equations

1. $x = y - 7$.
2. $x = \dfrac{y}{3}$.
3. $m = \dfrac{4 + 5n}{2}$.
4. $x = \dfrac{c - b}{a}$.
5. $q = \dfrac{3p - 12}{2}$; at $p = 6$: $q = \dfrac{18 - 12}{2} = 3$.
6. $b = \dfrac{20 - 5a}{2}$; at $a = 2$: $b = \dfrac{20 - 10}{2} = 5$.

Word Problems

1. $r = \dfrac{C}{2\pi} = \dfrac{18\pi}{2\pi} = 9$ ft.
2. $w = \dfrac{A}{\ell} = \dfrac{24}{6} = 4$ ft.
3. $r = \dfrac{d}{t} = \dfrac{330}{5.5} = 60$ mph.
4. (a) $F = \frac{9}{5}C + 32$. (b) $F = \frac{9}{5}(190) + 32 = 342 + 32 = 374°\text{F}$.

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